

Lab 4: A FPGA Calculator

This Lab is due 11:59PM May 24th. Your lab report and source code must be submitted by the deadline. This is the last lab of the class. Due to the university's deadline for submitting your scores, no extension of this lab is possible.

The lab includes a presentation before the deadline. You can record a video for the presentation, or make an appointment to present in the instructor's office.

This lab is to be done in teams. The required format for lab reports can be found on the resource page.

Objective: The objective of this laboratory is to implement a calculator with decimal or 2's complement display.

Equipment and software: You can use any FPGA boards of your choice for this lab. If you are going to use the Xilinx boards, you can download the Xilinx Vivado Development Environment for free from the Xilinx website.

Specification of the calculator: This lab is going to use 8 switches and 1 button on the FPGA board. You should implement the 8-bit ISA you designed in the first lab on the board. The 8-bit instruction words should be mapped to the 8 switches, and you should use one of the buttons for clock signal. Remember that your ISA is required to have a display instruction. For that instruction, you can choose to (1) display the content on the 4-digit seven-segment display on the board, or (2) use 8 on-board leds to display the values in 2's complement form. The method (2) is easier.

If you choose to display on the seven-segment display, all numbers are displayed **in decimal format** on the 4-digit seven-segment display on the board, even though the operands are read from the switches or registers as binary numbers. For example, if the content to display is "0101", the 4-digit seven-segment display should read as "[empty] [empty] [empty] 5".

Note that you may first need to implement a button debouncer that will help your calculator implementation, if you find the buttons on your board are particularly bouncy. The button debouncer is not required.

What to Turn In

For this lab project, turn in all of your source code and the code that tests your implementations. For each problem, all the source files and the project configuration files used by Vivado must be packed as a single zip file. The goal is that I can unpack your zip file and directly load your project into Xilinx Vivado. In your project report, please also describe your testing methodology.

For the presentation, each team is expected to showcase their calculator in a 5-minute presentation. Please generate the binary file before the presentation so that it can be directly transferred to your FPGA board and no time is wasted on synthesizing VHDL code. You are also expected to briefly describe your circuit schema, in particular the logic that drives the print instruction to either the seven-segment display or to on-board leds.

The presentation can be done on zoom or by sending a recording of your presentation to the instructor. The deadline of the presentation is the same as the lab deadline.

Peer evaluation: Please email your peer evaluation (guideline is on the resource page) of your teammates, including scores and justification, directly to the TA.